

## Solution Requirements

|                      |                                            |
|----------------------|--------------------------------------------|
| <b>Date</b>          | 06 July 2026                               |
| <b>Team ID</b>       |                                            |
| <b>Project Name</b>  | Voice Based Concept Understanding Analyser |
| <b>Maximum Marks</b> | 4 Marks                                    |

### Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

### Step 1: Functional Requirements (FR)

| S.No | Requirement Category | Requirement Description                           | Priority (High/Medium/Low) |
|------|----------------------|---------------------------------------------------|----------------------------|
| 1    | Authentication       | <i>User login using email/password</i>            | High                       |
| 2    | Authorization levels | <i>Student access to upload and view feedback</i> | Medium                     |
| 3    | External interfaces  | <i>Speech-to-text API (Google API)</i>            | High                       |

|   |                                   |                                                |        |
|---|-----------------------------------|------------------------------------------------|--------|
|   |                                   |                                                |        |
| 4 | Transactions processing           | <i>Process voice input and generate output</i> | High   |
| 5 | Reporting                         | <i>Show feedback and analysis results</i>      | Medium |
| 6 | Business rules, etc.              | <i>Analyze clarity, keywords, correctness</i>  | High   |
| 7 | Compliance to laws or regulations | <i>Ensure user data privacy</i>                | Medium |
| 8 | <i>Other</i>                      | <i>Store previous analysis results</i>         | Low    |

**Step 2: Non-Functional Requirements (NFR)**

| S.No | NFR Category | Requirement Description | Target Metric / Acceptance Criteria |
|------|--------------|-------------------------|-------------------------------------|
|      |              |                         |                                     |

|          |                                       |                                           |                                                                                  |
|----------|---------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------|
| <b>1</b> | <b>Performance &amp; Speed</b>        | <i>Response within 2–3 seconds</i>        | <i>System generates feedback within 3 seconds for voice input up to 1 minute</i> |
| <b>2</b> | <b>Scalability</b>                    | <i>Support multiple users</i>             | <i>System supports at least 50 concurrent users without performance drop</i>     |
| <b>3</b> | <b>Security &amp; Data Privacy</b>    | <i>Secure user data and voice input</i>   | <i>User data is encrypted and voice data is not stored without user consent</i>  |
| <b>4</b> | <b>Reliability &amp; Availability</b> | <i>Available 24/7 with minimal errors</i> | <i>System maintains at least 95% uptime during operation</i>                     |
| <b>5</b> | <b>Usability &amp; Accessibility</b>  | <i>Simple and user-friendly interface</i> | <i>Users can complete voice input and receive feedback without training</i>      |
| <b>6</b> | <b>Other</b>                          | <i>Easy to maintain and update</i>        | <i>System updates can be applied without affecting existing functionality</i>    |